Vaddi Karthik

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.89
Sri Chaitanya Junior college , Srikakulam	Class XII (Board of intermediate education, Andhra Pradesh India)	2020	9.94%
Ravindra Bharathi School, Srikakulam	Class X (Board of Secondary Education, Andhra Pradesh, India)	2018	10.00%

ACHIEVEMENTS

- **Completed a Minor Degree in Electronics and Communication Engineering** alongside the B.Tech. in Electrical Engineering at NIT Rourkela.
- **Awarded the Foundation for Excellence (FFE) Scholarship** throughout B.Tech. based on academic merit

MASTERS PROJECT

Hardware Quantization IP Blocks and FPGA Verification for YOLO V26

- Conducted a literature survey of **YOLO V26** and analyzed hardware-oriented quantization requirements for efficient neural-network inference.
- Designed and verified **asymmetric quantization (INT32 to FXP16)** and **requantization (FXP16 to INT8)** blocks using Python golden models and RTL implementations.
- Developed **RTL and SystemVerilog testbenches** for functional verification and evaluated the implemented IP blocks through FPGA-based testing.
- Performed **FPGA verification of a Depthwise Convolution IP** developed by a team member on **Xilinx ZCU104 using PetaLinux**, validating **APB control, AXI-Stream data transfer, and hardware outputs** against expected results.
- **Currently developing a generic PetaLinux-based FPGA IP verification framework** for input transfer, IP control, output capture, and hardware-versus-expected output comparison.

PROJECTS

Design and FPGA validation of Symmetric Quantisation & Dequantisation IPs on PYNQ-Z2| Verilog, AXI-Stream, Vitis

- Designed and verified **RTL-based symmetric quantisation and dequantisation blocks**, converting 16-bit fixed-point data to INT8 and reconstructing the approximate fixed-point values, with optimized RTL achieving **5000/5000 test cases passed**.
- **Integrated the** quantisation/dequantisation IP **with AXI-Stream and AXI DMA for FPGA testing**, developed Vitis software for data transfer and output verification, and validated the hardware using on-board ILA checks.

RTL-to-GDSII Implementation of 32-bit ALU using TSMC 180nm Technology | Cadence Xcelium, Genus, Innovus

- Designed and verified a **32-bit ALU supporting 16 operations** in Verilog; performed synthesis using Genus and achieved **70/70 PASS in Conformal LEC**.
- Implemented the complete **RTL-to-GDSII flow** in Innovus, including floorplanning, power planning, placement, CTS, routing and signoff, generating the final **GDSII layout in TSMC 180 nm**.

Design and Simulation of a 16×16 6T SRAM Array using Cadence Virtuoso| Cadence Virtuoso

- Designed and simulated a **16×16 SRAM array using 6T SRAM cells**, including precharge, write-driver, and sense-amplifier circuits.
- Designed the **row decoder and peripheral circuitry** and integrated them with the memory array for read/write

operation.

Design and Simulation of a Two-Stage CMOS OTA using GPDK 90nm

- Designed and simulated a **two-stage CMOS OTA** in Cadence Virtuoso with differential pair, active load, current-mirror biasing, and Miller compensation.
- Used **2 V supply, 3 µA bias current, 26 fF compensation, and 20 fF load**, achieving **68.42 dB nominal gain**.
- Performed **DC, AC, and PVT analysis** from **40°C to 125°C**, achieving **~70 dB maximum gain** with stable operation across process corners.

SKILLS

Languages: Verilog, C, Python.

Skills: ASIC Design, Physical Design, FPGA Prototyping, Hardware Software Interface

Protocols: AXI Lite, AXI Stream, UART

Tools: Xilinx Vivado, Vitis, Cadence Virtuoso, Cadence Genus, LtSpice.

RELEVANT COURSES

Digital Domain: Digital IC design, CMOS Digital Design, Introduction to HDL, Memory Device Technology for AI/ML Computing, Digital IC design Lab

Analog domain: Analog Electronics, Analog IC design.

Technology & devices: VLSI Technology, Semiconductor device modeling, Nano electronics: principles & devices.

POSITIONS OF RESPONSIBILITY

- Teaching assistant for Semiconductor devices and circuits lab by **Dr. Gajendranath Chaudhury**
- Teaching assistant for Probability and Random Processes course by **Prof. Anjana AM**

EXTRACURRICULAR

Hobbies: Badminton, Chess and Carrom